Republic of the Philippines
Western Mindanao State University
College of Computing Studies
DEPARTMENT OF COMPUTER SCIENCE

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Recommending approval:	**Ms. Lucy Felix-Sadiwa, MSCS**
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OUTCOMES-BASED EDUCATION (OBE) RESTRUCTURED COURSE SYLLABUS IN WEB DEVELOPMENT 1

First Semester, 2023-2024
Prerequisite: CC 101 Computer Programming 1

Western Mindanao State University

VISION

By 2040, WMSU is a Smart Research University generating competent professionals and global citizens engendered by knowledge from sciences and liberal education, empowering communities, promoting peace, harmony, and cultural diversity.

MISSION

WMSU commits to create a vibrant atmosphere of learning where science, technology, innovation, research, the arts and humanities, and community engagement flourish, and produce world-class professionals committed to sustainable development and peace.

College of Computing Studies

GOALS

The college shall provide academic excellence in the field of Information and Communication Technology, with emphasis on the following goals:

a. Produce quality, excellent and eco-friendly graduates imbued with gender responsiveness.
b. Achievement of the highest level of accreditation and center of excellence imbued with outcomes-based education.
c. Partner with national and international industries as an outlet for research development and extension.
d. Support faculty members through faculty development programs to be competitive with the highest global standards.

Bachelor of Science in Computer Science Program Outcomes	**GOALS**			
	A	B	C	D
A. Utilize effectively the concepts of computer science theories and methodologies and adopt new technologies and ideas in formulating effective solutions to address public health and safety, cultural, societal, gender, and environmental considerations. **(Competent Computer Science Professional, Socially Responsive)**	✓	✓	✓	

Bachelor of Science in Computer Science Program Outcomes	GOALS			
	A	B	C	D
B. Work cohesively with members of a team using their skills to successfully complete a project. **(Team Player)**	✓	✓	✓	
C. Pursue personal development and lifelong learning through research, graduate studies, training, and membership to a professional organization to be globally competitive. **(Lifelong Learner)**	✓	✓	✓	
D. Communicate effectively with the computing community and society through oral and written correspondence. **(Effective Communicator)**	✓	✓	✓	

COURSE CODE WD 114
COURSE NAME Web Development
PREREQUISITE CC 101 Computer Programming 1
COURSE CREDIT 2 Units Lecture/1 Unit Laboratory (5hrs/week)
COURSE DESCRIPTION This course covers the fundamental principles and technologies of web development. Students will learn HTML, CSS, and JavaScript basics, along with popular frameworks like Bootstrap. Through hands-on labs and projects, students will build responsive web pages and gain practical skills for creating dynamic web content. By the end of the course, students will be equipped with the essential knowledge to design and develop modern websites.

Course Learning Outcomes	Program Outcomes			
	A	B	C	D
A. Demonstrate proficiency in HTML, CSS, and JavaScript, including the ability to create well-structured and styled web pages.	✓	✓	✓	✓
B. Utilize CSS frameworks such as Bootstrap to efficiently design responsive and visually appealing layouts.	✓	✓	✓	✓
C. Implement form validation techniques to enhance the user experience and ensure data integrity.	✓	✓	✓	✓
D. Understand the Document Object Model (DOM) and utilize JavaScript to dynamically manipulate web page content.	✓	✓	✓	✓
E. Apply fundamental JavaScript programming concepts such as variables, conditions, functions, and loops in the context of web development.	✓	✓	✓	✓
F. Develop interactive web interfaces that respond to user actions and provide a seamless browsing experience.	✓	✓	✓	✓
G. Demonstrate the ability to collaborate effectively in a team setting, particularly in the context of project planning and execution.	✓	✓	✓	✓
H. Apply accessibility principles to ensure that web content is usable by individuals with diverse abilities and disabilities.	✓	✓	✓	✓

Course Learning Outcomes	Program Outcomes			
	A	B	C	D
I. Evaluate and utilize various web development tools and resources to streamline the development process and troubleshoot issues effectively.	✓	✓	✓	✓
J. Present and communicate web development projects effectively, showcasing understanding of design principles and technical implementation.	✓	✓	✓	✓

References	1. "HTML and CSS: Design and Build Websites" by Jon Duckett 2. "CSS Mastery: Advanced Web Standards Solutions" by Andy Budd, Cameron Moll, and Simon Collison 3. "JavaScript: The Good Parts" by Douglas Crockford 4. "Learning Responsive Web Design: A Beginner's Guide" by Clarissa Peterson

Grading System

Rating Criteria

Attendance – 5%
Lab Activities/Seat Work – 15%
Quizzes – 20%
Major Exam – 30%
Project – 30%

Passing: 60%

Sample computation:
(Score/total item) X 100
44/50 X 100
88.0 or 1.5

% EQUIVALENT	NUMERICAL RATING
96-100	1.0
91-95	1.25
86-90	1.5
81-85	1.75
76-80	2.0
71-75	2.25
66-70	2.5
62-65	2.75
60-61	3.0
Below 60/Failed the Project Defense	5.0
Lacks requirements/no final exam/no presentation	INC
Authorized Withdrawal (Dropped with permit)	AW
Unauthorized Withdrawal (Dropped from class for non-attendance/non-appearance for 20% of prescribed attendance)	UW

Course Requirements

- Attendance
- Lab Activities
- Quizzes
- Major Exam
- Project

Classroom Policy
Attendance: Per article 286 of the WMSU code, regular attendance is required of all students. Attendance is counted upon the first day of regular classes, regardless of the time of the student's enrolment. Students who came into class within 15 minutes after the start time will still be considered present; otherwise, the student will be marked as late. Seven consecutive absences may be a ground for students to be dropped in the subject. Students who cannot attend either face-to-face or online class due to illness must contact the instructor immediately regarding the absence and discuss any activities missed in the class. **Wearing of Prescribed School Uniform and ID:** Per article 391 of the WMSU code, every student must wear the prescribed college uniform unless, for certain valid reasons, s/he has written exemption from the Dean of Student Affairs which s/he must show on demand. Students are exempted from wearing their school uniforms during Wednesdays and Saturdays since these days are considered a wash day. Students are also required to wear their school ID when inside the campus, at all times. For online classes via video conferences, students are not required to wear their school uniform and advise to wear any appropriate attire. **Cleanliness and Room Organization:** For lecture and laboratory, students are advised to arrange the chairs and tables (if any), and pick up pieces of trash before the start of the class. A student is also assigned at random to keep the white or blackboard clean before and after the class. Lighting, air conditioners, and other electrical equipment should also be turned on and off during and after the class respectively. An additional rule for the laboratory is that students are required to wear their shoe protectors when coming into the laboratory room. During online classes, students are expected to attend their classes in a quiet place. **Seat Plan:** For the seating plan during lecture classes, students are given the freedom to choose the seat and place they are comfortable with inside the room. For the laboratory, if the instructor deems it necessary, students are rearranged randomly. There should be one computer for each student. **Cellphone/Mobile Devices:** Cellular phones or any mobile device should be set to silent or vibrate mode to avoid disruption especially during face-to-face class discussion. Important calls and texts must be answered outside the classroom. Students are permitted to use their mobile phones when attending their online classes. The use of laptops/computers is recommended for a better learning experience.

Classroom Policy

Student Facilitators: Student facilitators are selected from the class by the teacher. These students will assist their classmates in accomplishing the laboratory activity given by the teacher. This is done only upon the instruction of the teacher and also applicable during online classes but not to the extent of sharing/copying answers.

Quizzes: Quizzes are given to students at least once after a chapter or two which should be announced. Students are advised to use the CET Exam Booklet during quizzes. For online quizzes/exams, students will take the quiz/exam through the Learning Management System platform, questions and options are shuffled, navigation is disabled if necessary. Quiz/exam is always time-pressured.

Major Examinations: Major exams such as midterm and final exams are given to students based on the schedule specified in the university calendar. They are also advised to use the CET examination booklet for the major exams. Students who fail to take the major exams due to illness (with a medical certificate) should inform the instructor early so they will be given a special exam before the submission of grades. Otherwise, an "INC with No Final/Midterm exam" will be submitted by the instructor. A calculator may be used depending on the content of the exam. However, mobile devices are strictly prohibited during major exams. Failure to do so will be considered a form of cheating. Major exams can be also administered online in the same manner how the online quizzes are conducted.

Laboratory Activities: Laboratory hands-on activities are given to students at every meeting. The laboratory output is graded based on a rubric scorecard. A major laboratory hands-on activity is given to students during the midterm and final exam week.

Consultation: Aside from the regular system of consultation offered by the adviser and/or guidance counselor, the faculty shall provide the student with consultation hours for students to inquire regarding their subject/course, ask for clarification or further explanation of certain topics discussed, or to be discussed in the subject, and other concerns related to the subject. The student should set an appointment at least a day before the consultation to avoid conflicts of schedule. Consultation can be done online through any messaging platform.

Student Integrity/Academic Honesty

Cheating

Receiving or providing unapproved help in any academic task, test, or treatise. Cheating includes the attempt to use or the actual use of any unauthorized information, educational material, or learning aid in a test or assignment. Cheating includes multiple submissions of any academic exercise more than once for credit without prior authorization and approval of the instructor.

Plagiarism

Classroom Policy

Presenting someone else's work as though it is your own. In an academic community, the use of words, ideas, or discoveries of another person without explicit, formal acknowledgment constitutes an act of theft or plagiarism. In order to avoid the charge of plagiarism, students must engage in standard academic practices such as paraphrasing or replacing words that are not their own and employing the appropriate documentation or citation, and including a formal acknowledgment of the source in the proper format.

Fabrication, Obstruction, and Collusion

Fabrication – Inventing or falsifying any data, information, or records.
Obstruction – Impeding the ability of another student to perform assigned work.
Collusion – Assisting any of the above situations or performing work that another student presents as his or her own.

Penalties

Penalties for an academic offense include one or more of the following:

- Resubmission of the work in question
- Submission of additional work for the course in which the offense occurred
- A lower grade or loss of credit for the work found to be in violation
- A failing grade of 5.0 for the course in which the offense occurred

Submission/Presentation of Projects/Requirements: Submission or presentation of projects or requirements must be on time. No project or requirement will be accepted after the due date unless with a valid reason.

Additional Online Course Policy

Learning Modality: Blended Learning Approach will be used if face-to-face classes are not possible. A mixture of synchronous virtual classes and asynchronous online class discussions are to be expected to support remote learning. 70% of the classes and/or meetings will be purely online including project presentations. Students who will be employing face-to-face group internal activities such as group meetings, collaboration, and alike should inform their instructor ahead of time. IATF protocols should be observed at all times when doing face-to-face activities.

The Learning Management System (LMS) Platform

- MS Teams - is used for online meetings, virtual classes, online presentations, and alike. Google Meet can still be used for the same purpose.

Classroom Policy

- Messenger - is used for informal class conversations such as instructions, announcements, etc.
- LMS (Digital Education) - the www.eclass.wmsu.edu.ph is used for e-learning activities. However, Quizzes can only be taken in the LMS during the class schedule. Assignments, activities, and all other requirements required by the instructor should be submitted using the LMS Digital Education platform.
- Important notes: in the effort of aiding remote learning in our institution, the LMS Digital Education Platform used in this course is for testing purposes only. Instructors of your other subjects are still using the officially endorsed LMS (Google Classroom, MS Teams).

Video Recording: Students are not allowed to record their online classes unless otherwise permitted by the instructor.

Due Dates and Late Work: Students are given at least 1 week to complete their given activity except for proctored quiz/exam. For late work, submission is subject to point deduction depending on the number of days late. Resubmission's highest grade is 80.

Inquiry and Grading Feedback Turn-Around Time: For inquiries, students shall expect responses from their instructor within 24-48 hours. Expect at least 2 weeks after submission for the grading and releasing of results.

Technical Issues and Temporary Fallback: Students shall inform their instructor if they failed to join online classes due to technical issues, internet connectivity problems, invalid accounts, electricity interruption, etc. In any case, the LMS is not available, other alternative platforms shall be used to continue e-learning.

Student Responsibilities: Students are expected to demonstrate the technical and academic skills to be a successful online student; one who can operate the computer and internet functions necessary to complete an online class and someone who has the self-discipline and time management skills to work independently. Though e-learning materials can be accessed using their mobile phones, students shall have ready access to a computer and the internet and a backup plan if hardware or internet access fails. Students should use software that complies with the minimum requirements of software necessary to access and run the course, obtain required textbooks, and if needed, related course materials or supplies. Students are expected to log in and contribute to the online class regularly in a meaningful way. They should also know and comply with the due date and deadline requirements of the course and communicate with the instructor and fellow students in a professional, polite manner, using appropriate tone, language, grammar, and spelling. They should also easily locate and use academic support resources such as tutoring, writing consultation, and library services, demonstrate commitment to academic honesty by completing assignments and other coursework with integrity, monitor and ask questions about their course grade, save backup copies of lectures, assignments, and class emails and comply with online participation requirements to avoid being dropped for non-participation.

Course Rubrics

Lab Activity					
Criteria	**Excellent (4)**	**Good (3)**	**Satisfactory (2)**	**Needs Improvement (1)**	**Unsatisfactory (0)**
Technical Proficiency (20%)	Student demonstrates exceptional mastery of web development concepts and tools, producing error-free and efficient code.	Student displays a solid understanding of web development concepts and effectively applies tools, with minor errors or inefficiencies.	Student shows basic competency in web development but may encounter issues in implementation or execution.	Student's technical skills are lacking, resulting in multiple errors and a suboptimal final product.	Student's work indicates a lack of understanding or incorrect implementation of fundamental web development principles.
Problem Solving (20%)	Student identifies and addresses complex issues, using innovative solutions to overcome challenges effectively.	Student identifies and solves problems using appropriate strategies, with minor oversight or less originality.	Student attempts to troubleshoot and solve issues but may require guidance or miss important details.	Student struggles to identify and resolve problems, relying heavily on assistance or not effectively addressing issues.	Student demonstrates an inability to identify and address problems, leading to incomplete or non-functional work.
Creativity and Design (20%)	Student creates a visually appealing and user-friendly interface, showcasing creative and thoughtful design choices.	Student designs an aesthetically pleasing interface with logical user interactions and design considerations.	Student presents a functional design, though it may lack finesse or originality.	Student's design is basic and may not prioritize user experience or visual appeal.	Student's design lacks coherence, aesthetics, and consideration for user experience.
Code Organization and Quality (10%)	Student maintains impeccable code structure, naming conventions, and organization, enhancing code readability and maintainability.	Student adheres to good code practices, demonstrating organized and well-structured code with minimal inconsistencies.	Student's code is organized, though there may be occasional readability issues or minor deviations from best practices.	Student's code organization is lacking, leading to confusion and readability problems.	Student's code is disorganized, poorly structured, and difficult to understand or maintain.

Lab Activity					
Criteria	**Excellent (4)**	**Good (3)**	**Satisfactory (2)**	**Needs Improvement (1)**	**Unsatisfactory (0)**
Functionality and Features (20%)	Student fully implements all required functionality and incorporates advanced features that enhance user experience.	Student implements required functionality accurately and includes additional features, though some may be incomplete or less polished.	Student partially implements required functionality, omitting certain features or encountering issues in execution.	Student's implementation lacks significant functionality, resulting in incomplete or non-functional features.	Student's work demonstrates a severe lack of functional implementation, rendering key features unusable or absent.
Documentation and Comments (10%)	Student provides comprehensive and clear documentation, including comments that thoroughly explain code logic and decisions.	Student includes meaningful comments and documentation that aid in understanding code logic and structure.	Student includes minimal comments and documentation, providing some insight into the code but lacking detail.	Student's documentation is insufficient, making it challenging to understand the code's purpose or functionality.	Student's documentation is absent or incomplete, leaving no guidance for understanding the code.

Course Project					
Criteria	**Excellent (4)**	**Good (3)**	**Satisfactory (2)**	**Needs Improvement (1)**	**Unsatisfactory (0)**
Concept and Innovation	Student presents a highly innovative and original concept, demonstrating a deep understanding of user needs and industry trends.	Student proposes a creative and well-defined concept that addresses user needs and includes some innovative elements.	Student's concept is clear and relevant, though it may lack distinctiveness or originality.	Student's concept is basic and may not fully meet user needs or engage with current trends.	Student's concept lacks creativity and innovation, offering little value to users.
Functionality and Features	Student implements a comprehensive set of features, exceeding expectations and showcasing advanced functionality.	Student successfully implements required functionality and includes additional features that enhance the application's value.	Student implements core functionality, but some required features may be incomplete or have minor issues.	Student's implementation lacks significant functionality, leading to incomplete or non-functional features.	Student's application lacks essential functionality, rendering key features unusable or absent.

Course Project					
Criteria	**Excellent (4)**	**Good (3)**	**Satisfactory (2)**	**Needs Improvement (1)**	**Unsatisfactory (0)**
User Experience	Student delivers an exceptional user experience, with intuitive navigation, user-friendly interfaces, and seamless interactions.	Student provides a well-thought-out user experience, considering user needs and implementing effective design principles.	Student's application offers a satisfactory user experience, but there may be occasional usability issues or design inconsistencies.	Student's application has noticeable usability problems, hindering user interaction and experience.	Student's application provides a poor user experience, making navigation and interaction challenging for users.
Design and Aesthetics	Student creates an aesthetically pleasing and cohesive design that enhances the application's overall appeal and usability.	Student presents a visually appealing design with a consistent theme and layout, though some design elements may be less refined.	Student's design is acceptable but lacks cohesiveness, and certain design choices may not align well with the application's purpose.	Student's design is basic and may lack visual appeal, with inconsistencies that affect overall presentation.	Student's design is unpolished, lacks coherence, and fails to contribute positively to the application's aesthetics.
Code Quality and Structure	Student maintains impeccable code organization, structure, and naming conventions, ensuring readability and maintainability.	Student's code is well-organized and adheres to good practices, with minor inconsistencies that do not hinder readability.	Student's code structure is satisfactory, though there may be readability issues or deviations from best practices.	Student's code is disorganized or poorly structured, leading to confusion and difficulties in understanding.	Student's code is messy, lacks structure, and is challenging to follow or maintain.
Documentation and Deployment	Student provides comprehensive and well-structured documentation, including setup instructions and explanations for code logic. Deployment is smooth and successful.	Student includes clear documentation and deployment instructions, aiding in understanding and facilitating successful deployment.	Student's documentation is sufficient but may lack detail or clarity in certain areas. Deployment is functional, but there may be minor issues.	Student's documentation is lacking essential details, making it difficult to understand code logic or deployment steps. Deployment encounters significant problems.	Student's documentation is minimal or absent, and deployment fails to work properly.

Course Outline							
Time Frame	**Course Content (No. of Hours Per Topic)**	**Desired Student Learning Outcomes/Competencies At the end of each topic and semester, the students can**	**Outcome-Based (OBA) Activities (Teaching & Learning Activities)**	**Evidence of Outcomes (Assessment of Learning Outcome)**	**Course Learning Outcomes**	**Program Outcomes**	**Values Integration**
Week 1 WMSU VMGO and Course Orientation	4 hours ✓ WMSU VMGO, Classroom Policies, Course Overview, Course Requirements, Grading System	✓ Discuss the purposes of curriculum development and their relations to VMGO. ✓	✓ Role play how they will achieve the VMGO in the said course. ✓ Set the classroom policies by groups then come up with a consensus the classroom policies.	✓ Use a rubric score card to assess their quality of participation	A,F,G,H	A,B,C,D	✓ Appreciation ✓ Team player
Week 2 – 3 Understand the basic elements and structure of HTML documents	10 hours Elements and Structure: ✓ Basic HTML ✓ Document Structure ✓ Tags and Attributes Semantic HTML: ✓ Semantic	✓ Understand the fundamental structure of HTML documents. ✓ Identify and utilize HTML tags and attributes effectively. ✓ Demonstrate proficiency in creating well-structured HTML content.	✓ Lecture on basic HTML syntax and document structure. ✓ Hands-on exercises creating HTML documents with various elements. ✓ Group discussions on the importance	✓ Completion of HTML coding assignments demonstrating proper document structure and tag usage. ✓ Quizzes assessing understanding of HTML elements and attributes.	A,F,G,H	A,B,C,D	✓ Critical thinking and adaptability ✓ Ethical responsibility and professionalism ✓ Respect for diversity and inclusivity

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	Markup ✓ SEO & Accessibility ✓ HTML5 Elements		of proper HTML structure.	✓ Peer review of HTML documents for adherence to best practices. ✓ Lab Act 01: Food Blog			
WEEK 3 HTML Tables	4 Hours ✓ Introduction to HTML Tables ✓ Creating Basic Tables ✓ Table Attributes and Elements ✓ Styling Tables with CSS	✓ Understand the purpose and structure of HTML tables. ✓ Demonstrate proficiency in creating tables for organizing data. ✓ Apply advanced table formatting techniques using CSS. ✓ Ensure tables are accessible and optimized for performance.	✓ Lecture and discussion on HTML table fundamentals. ✓ Hands-on practice creating tables with varying attributes. ✓ Interactive demonstration of CSS styling for tables. ✓ Implementation of accessibility features in tables. ✓ Guided exercises on optimizing	✓ Completion of HTML table assignments demonstrating proficiency in table creation and styling. ✓ Peer review of table designs assessing clarity and organization. ✓ Presentation of creatively styled and accessible tables using CSS and semantic markup.	A,B,F,G,H	A,B,C,D	✓ Emphasis on clear and organized presentation of data. ✓ Encouragement of creativity and innovation in table design. ✓ Collaboration and feedback exchange during peer evaluation activities. ✓ Respect for inclusivity and accessibility in web design.

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			table structure and performance.	✓ Lab Act 02: Formatting Tables			
Week 5 JavaScript and jQuery	6 Hours ✓ DOM Manipulation with JavaScript ✓ Event Handling and Event Listeners ✓ DOM Manipulation with jQuery ✓ Building Responsive and Interactive UIs	✓ Modify webpage content and structure dynamically using JavaScript's Document Object Model (DOM) manipulation techniques. ✓ Apply DOM traversal methods to select and manipulate specific elements within the HTML document. ✓ Implement event listeners to respond to user interactions effectively. ✓ Create responsive web interfaces by associating JavaScript functions with specific events. ✓ Utilize jQuery library functions to	✓ Students create a webpage where they use JavaScript to dynamically modify content based on user input or button clicks. ✓ Students enhance a webpage by implementing event listeners and handlers to create responsive interactions, such as toggling elements or displaying alerts. ✓ Students rewrite a	✓ Submitted webpage where JavaScript effectively modifies content in response to user actions, demonstrated through interactive elements or dynamic changes. ✓ Screenshots or code samples showing event listeners and handlers in action, highlighting	A,B,C,D	A,B,C,D	✓ User-Centered Approach ✓ Effective Communication ✓ Efficiency and Collaboration ✓ Creativity and User-Centric Design

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		streamline DOM manipulation tasks. ✓ Employ jQuery methods to achieve similar outcomes as traditional JavaScript DOM manipulation. ✓ Develop interactive user interfaces (UIs) that respond to user actions and provide real-time feedback. ✓ Create responsive UI components using DOM manipulation and jQuery for enhanced user experience.	portion of a webpage's DOM manipulation using jQuery, showcasing familiarity with jQuery's concise syntax. ✓ Students design and build a responsive UI component (e.g., a collapsible navigation menu) that engages users through event-driven interactivity.	how user interactions trigger responsive behaviors. ✓ Side-by-side comparison of code snippets, demonstrating the student's ability to achieve similar DOM manipulation outcomes using both JavaScript and jQuery. ✓ Working interactive UI component (e.g., collapsible menu) hosted online, along with the			

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				associated HTML, CSS, and JavaScript code. ✓ Rubrics for different activity.			
Week 6 – 8 Web Programming with PHP and MySQL Part 1	18 Hours ✓ Introduction to PHP and MySQL ✓ Setting Up Development Environment ✓ PHP Syntax and Variables ✓ Data Types and Operators in PHP ✓ Working with Strings and Arrays ✓ Control Structures: Conditionals and Loops ✓ Functions and	✓ Understand the role of PHP and MySQL in web development. ✓ Explain the benefits of using PHP and MySQL for building dynamic web applications. ✓ Configure a functional development environment for PHP and MySQL. ✓ Install and set up necessary tools and software for web development. ✓ Write PHP code using correct syntax, including variables and data types.	✓ Students research and present on real-world examples of dynamic websites powered by PHP and MySQL. ✓ Students follow step-by-step instructions to set up a local development environment for PHP and MySQL. ✓ Students complete coding	✓ Presentation slides or written reports highlighting real-world examples of dynamic websites, showcasing understanding of PHP and MySQL's role. ✓ Screenshots or documentation showing a functional local development	E,F,G,H,I,J	A,B,C,D	✓ Collaboration and Teamwork ✓ Efficiency and Sustainability ✓ Attention to Detail ✓ Problem Solving ✓ Diversity and Inclusion ✓ Adaptability and Persistence ✓ Attention to Quality ✓ Ethical User-Centered Design ✓ Responsibility and Integrity

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	Scope in PHP ✓ Form Handling and Data Validation ✓ Introduction to Databases and MySQL ✓ Creating and Managing Databases and Tables ✓ SQL Queries: SELECT, INSERT, UPDATE, DELETE ✓ Data Retrieval with PHP and MySQL ✓ Data Manipulation with PHP and MySQL	✓ Demonstrate an understanding of PHP's fundamental programming concepts. ✓ Utilize various data types and operators in PHP to perform calculations and manipulate data. ✓ Apply appropriate operators for string manipulation, arithmetic, and comparisons. ✓ Manipulate strings and arrays effectively using PHP functions and methods. ✓ Implement operations such as concatenation, substring extraction, and array sorting. ✓ Implement control structures, including conditionals (if, else) and loops (for, while).	exercises to practice PHP syntax and variable usage. ✓ Students create PHP scripts to perform data manipulations using various data types and operators. ✓ Students work on hands-on exercises involving string and array operations in PHP. ✓ Students solve coding challenges that require implementing conditional statements and loops.	environment for PHP and MySQL. ✓ Quiz results or code snippets demonstrating correct usage of PHP syntax and variables. ✓ Submitted PHP scripts performing data manipulations using various data types and operators. ✓ Completed lab exercises showcasing manipulation of strings and arrays in PHP.			✓ Organization and Order ✓ User-Centric Approach ✓ Ethical Data Handling

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		✓ Construct logical flow in PHP programs using conditional statements. ✓ Create functions with appropriate parameters, return values, and variable scope. ✓ Understand the concept of scope and apply it in function definitions. ✓ Develop forms in HTML and handle form submissions using PHP. ✓ Implement data validation techniques to ensure secure and accurate input. ✓ Define databases and explain their importance in web applications. ✓ Describe the role of MySQL as a relational	✓ Students design and implement functions to solve specific programming tasks. ✓ Students create a form that handles user input, validates data, and provides appropriate feedback. ✓ Students participate in a group discussion on the importance and uses of databases in web applications. ✓ Students design a normalized database schema for a	✓ Code solutions or written explanations demonstrating correct implementation of conditionals and loops. ✓ Submitted functions with appropriate parameters, return values, and variable scope. ✓ Submitted form handling PHP scripts, demonstrating proper data validation and feedback. ✓ Participation in the group			

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		database management system. ✓ Design and create databases and tables in MySQL, adhering to normalization principles. ✓ Understand the structure of tables, columns, and data types. ✓ Write SQL queries for selecting, inserting, updating, and deleting data in MySQL. ✓ Understand the syntax and usage of basic SQL statements. ✓ Retrieve data from MySQL databases using PHP. ✓ Display retrieved data on web pages and format it appropriately.	given scenario and create corresponding tables in MySQL. ✓ Students write and execute SQL queries to retrieve, insert, update, and delete data in a database. ✓ Students retrieve data from a MySQL database using PHP and display it on a webpage. ✓ Students create PHP scripts to perform data manipulation operations (update, insert, delete) in a	discussion and contributions to the conversation on database importance and usage. ✓ Submitted normalized database schema and table creation scripts for a given scenario. ✓ Executed SQL queries and corresponding output demonstrating correct usage of query statements. ✓ Functional webpage			

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		✓ Update, insert, and delete data in MySQL using PHP scripts. ✓ Ensure data consistency and integrity when performing data manipulation operations.	MySQL database.	displaying data retrieved from a MySQL database using PHP. ✓ Submitted PHP scripts successfully performing data manipulation operations (update, insert, delete) in a MySQL database. ✓ Rubrics for different activity. ✓ Quiz 02			
Week 9	MAJOR EXAM (WRITTEN & HANDS-ON)						
Week 10 – 17 Web Programming	48 Hours ✓ Handling User	✓ Implement secure user authentication mechanisms to protect	✓ Students implement a secure user	✓ Submitted PHP code	E,F,G,H,I,J	A,B,C,D	✓ Ethical Responsibility

Course Outline							
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with PHP and MySQL Part 2	Authentication and Security ✓ Sessions and Cookies in PHP ✓ File Handling and Uploading ✓ Accessible Web Development Principles ✓ Implementing Web Accessibility Standards ✓ Building Accessible Forms and User Interfaces ✓ Deployment Strategies and Web Hosting ✓ Version Control with Git and GitHub ✓ Introduction to MVC Architecture ✓ Building a	sensitive data and user accounts. ✓ Utilize sessions and cookies to maintain user state and enhance user experience. ✓ Develop web applications capable of securely handling file uploads and processing. ✓ Understand the importance of accessible web development and its impact on user inclusivity. ✓ Apply best practices to create web content that's accessible to users with disabilities. ✓ Create forms and interfaces that meet accessibility guidelines, ensuring usability for all users.	registration and login system, incorporating encryption and validation techniques. ✓ Students create a PHP application that uses sessions and cookies to maintain user preferences across different pages. ✓ Students build a web application that allows users to securely upload files, verifying and processing them as needed. ✓ Students evaluate an existing	showcasing a functional and secure user registration and login system with encryption and validation. ✓ Screen recordings or code snippets demonstrating the creation and usage of sessions and cookies in a PHP application. ✓ Functional web application that securely uploads, verifies, and processes files,			✓ Respect for User Preferences ✓ User-Centered Design ✓ Inclusivity and Empathy ✓ Respect for User Diversity ✓ Equal Access and Usability ✓ Effective Planning and Resource Management ✓ Collaboration and Teamwork ✓ Clarity and Structured Design ✓ Creativity and Problem Solving ✓ User-Centric Design ✓ Attention to Quality and

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	Simple PHP and MySQL Web Application ✓ Web Performance Optimization ✓ Testing and Debugging Web Applications ✓ Responsive Design and Mobile Compatibility ✓ Web Security Best Practices ✓ Content Management Systems (CMS) and PHP ✓ Integrating Third-Party APIs ✓ Web Application Deployment and Hosting	✓ Plan and execute effective deployment strategies for hosting PHP and MySQL web applications. ✓ Employ version control using Git and GitHub for collaborative and efficient code management. ✓ Comprehend the Model-View-Controller (MVC) architecture and its role in building scalable web applications. ✓ Develop a functional web application that integrates PHP and MySQL, showcasing practical skills. ✓ Implement techniques to optimize web application performance,	website for accessibility issues and propose improvements based on accessibility principles. ✓ Students redesign a webpage to meet accessibility standards, ensuring it's usable for users with disabilities. ✓ Students design and create an accessible web form that accommodates users of all abilities. ✓ Students outline a step-	accompanied by the relevant PHP code. ✓ Accessibility audit report highlighting identified issues and proposed improvements for a given website. ✓ Before-and-after screenshots or code examples showing the redesigned webpage following accessibility standards. ✓ Submitted accessible web			Continuous Improvement ✓ Adaptability and User-Centered Design ✓ Integrity and Trust ✓ Adaptability and Integration ✓ Collaboration and Openness ✓ Responsibility and Accountability

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		enhancing user experience. ✓ Apply debugging strategies and testing methodologies to ensure robust and error-free web applications. ✓ Apply security practices to safeguard web applications from vulnerabilities and threats. ✓ Integrate PHP with content management systems to create dynamic and feature-rich websites. ✓ Integrate external APIs to enhance functionality and data exchange within web applications. ✓ Successfully deploy and host web applications on various platforms,	by-step deployment plan for hosting a PHP and MySQL web application on a chosen platform. ✓ Students collaborate on a group project using Git and GitHub, demonstrating effective version control practices. ✓ Students design a simple web application following the MVC architecture, outlining the responsibilities of each component.	form with well-structured HTML and appropriate attributes for different user abilities. ✓ Written deployment plan outlining steps, configurations, and considerations for hosting a PHP and MySQL web application. ✓ GitHub repository showcasing collaborative coding, version history, and			

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		ensuring accessibility and availability.	✓ Students create a functional PHP and MySQL web application that incorporates user authentication, database interactions, and user-friendly interfaces. ✓ Students analyze and optimize the performance of a web page, implementing techniques to improve loading speed and responsiveness. ✓ Students identify and fix bugs in a	clear commit messages. ✓ Diagram or documentation detailing the model, view, and controller components of a designed web application. ✓ Fully functional web application hosted online, demonstrating user authentication, database interactions, and responsive design. ✓ Performance analysis report and modified webpage			

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			provided PHP application, applying debugging techniques and writing test cases. ✓ Students perform a security assessment on a web application, identifying vulnerabilities and proposing security measures. ✓ Students integrate PHP functionality into a chosen content management system, enhancing	showing improved loading speed and responsiveness. ✓ Bug report with identified issues, corresponding fixes, and test cases demonstrating successful debugging. ✓ Security assessment report detailing vulnerabilities found, along with proposed security measures. ✓ Integrated PHP functionality			

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			website capabilities. ✓ Students integrate a third-party API into a web application, demonstrating data exchange and integration skills. ✓ Students simulate the deployment and hosting process of a PHP and MySQL web application on a local server or cloud platform.	within a CMS, showcasing expanded capabilities and functionality. ✓ Functional web application integrating a third-party API and demonstrating data exchange and usage. ✓ Simulation report outlining deployment steps, configurations, and a locally/online hosted PHP and MySQL web application.			

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				✓ Rubrics for different activity. ✓ Quiz 03 and 04			
Week 18	**PROJECT PRESENTATION** **Developing a Functional Web Application Hosted Online** In this project, students will create a functional web application that integrates various concepts covered in the course. The web application will include data manipulation, user authentication, database interactions, responsive design, web accessibility, and deployment. Students may use frameworks and templates to speed up development. Deployment is also left to the discretion of the student.						

Prepared By:	**Noted By:**	**Recommending Approval By:**	**Approved By:**
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